

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [172, 132, 151, 129, 133, 125, 137, 116, 109, 134]

Tap amplitudes (pixels): [161.4465855941215, 183.89671013914307, 188.50198937942272, 199.80490484470096, 196.31097778779463, 190.90573590125572, 191.34262462922368, 179.4017837146554, 176.08236708995025, 172.74258305351347, 204.70710783946902]

GPT-5

MDS-UPDRS Item 3.4: Finger Tapping — Scoring Analysis

What the data show

- Declared instruction: 10 taps; data include 11 amplitudes and 10 inter-tap periods (which is typical: N amplitudes often accompany N or N+1 detected peaks). I'll evaluate the first 10 taps to align with the instruction.
- Tap periods (frames, first 10): [172, 132, 151, 129, 133, 125, 137, 116, 109, 134]
 - Trend: After the first tap, periods generally shorten toward the middle (e.g., 172 → 109 by tap 9), then slightly increase at the last interval (134). This suggests an initial settling then relatively steady pace with mild variability. No obvious repeated hesitations or long arrests.
- Tap amplitudes (pixels, first 10): [161.45, 183.90, 188.50, 199.80, 196.31, 190.91, 191.34, 179.40, 176.08, 172.74]
 - Pattern: Amplitude increases from tap 1 to a peak around taps 3–4, then shows a gradual, monotonic-like decline from tap 4 onward through tap 10.
 - Decrement timing: The amplitude decrement begins early—effectively after the 1st–2nd taps, it rises briefly, peaks by tap 3–4, and then consistently declines through the remainder. This is more than just an “end-only” decrement; it’s present across the latter majority of the sequence.

Mapping to scoring criteria

- Interruptions/hesitations: None clearly indicated (>5 interruptions or a long arrest not evident). Periods are variable but do not show clear arrests.
- Speed: Periods suggest only mild-to-moderate variability; no marked slowing throughout, and actually some speeding early. Not consistent with “moderate slowing.”
- Amplitude decrement:
 - Not limited to the very end (which would support 1).
 - The decline is evident well before the end—essentially midway and persisting to the finish, aligning with “decrements midway” (supports 2).
 - “Decrements starting after the 1st tap” (criterion for 3) typically implies an immediate and persistent decrement right from the second tap forward without the early increase seen here. Here, amplitude increases through taps ~3–4 before declining. Thus it does not cleanly meet the “starting after 1st tap” pattern for a score of 3.

Conclusion

- Best fit: 2 (Mild), based on amplitude decrement becoming established in the mid-sequence, without clear multiple interruptions or moderate slowing.

Score: 2
